

Summary and Recommendation

1. Project Objective

The objective of this analysis was to identify the factors influencing customer churn in a telecommunications company and derive actionable business insights that can help improve customer retention. Customer churn refers to customers discontinuing their services with the company. Understanding churn behavior is critical because acquiring new customers is often more expensive than retaining existing ones.

2. Data Understanding and Preparation

The dataset contains customer demographic information, subscription details, contract information, billing details, and churn status.

Data Cleaning Performed

- Checked dataset dimensions, data types, and overall structure.
- Identified missing values in the **TotalCharges** column.
- Replaced blank values in **TotalCharges** with 0 and converted the column to numeric format.
- Verified that no duplicate customer records existed.
- Converted **SeniorCitizen** values from numerical format (0 and 1) to categorical labels (No and Yes) to improve interpretability during analysis.

The dataset was successfully cleaned and prepared for exploratory data analysis (EDA).

3. Exploratory Data Analysis (EDA)

Customer Churn Distribution

The first analysis focused on understanding the overall churn distribution.

Findings:

- The majority of customers remained with the company.
- Approximately 25–30% of customers had churned.

- This indicates that while customer retention is generally good, a significant proportion of customers are still leaving the service.

Business Insight:

Reducing churn even by a few percentage points could result in substantial revenue retention.

Gender-wise Churn Analysis

Customer churn was analyzed across male and female customers.

Findings:

- Male and female customers exhibited very similar churn behavior.
- No major differences were observed in churn rates between genders.

Business Insight:

Gender does not appear to be a strong predictor of churn and may not require targeted retention campaigns.

Senior Citizen Analysis

The relationship between senior citizens and churn was investigated.

Findings:

- Senior citizens showed a higher tendency to churn compared to non-senior customers.
- A greater proportion of senior customers left the service.

Business Insight:

Senior citizens may require customized service plans, simplified billing, or dedicated customer support to improve retention.

Tenure Analysis

Tenure represents the number of months a customer has remained with the company.

Findings:

- Customers with very low tenure (new customers) showed the highest churn.
- Customers with longer tenure generally remained loyal to the company.
- Churn decreases as customer tenure increases.

Business Insight:

The first few months are critical in the customer lifecycle. Companies should focus retention efforts on newly acquired customers through onboarding programs and promotional offers.

Contract Type Analysis

Customers were categorized according to their contract type:

- Month-to-Month
- One-Year
- Two-Year

Findings:

- Month-to-month customers exhibited the highest churn rate.
- Customers with one-year contracts churned significantly less.
- Two-year contract customers showed the lowest churn rate.

Business Insight:

Long-term contracts create customer commitment and reduce churn risk. Encouraging customers to move from month-to-month plans to annual plans can improve retention.

Payment Method Analysis

Different payment methods were analyzed for their relationship with churn.

Findings:

- Customers using Electronic Check showed the highest churn levels.
- Customers using automatic payment methods demonstrated better retention.

Business Insight:

Electronic check users may experience payment friction or service dissatisfaction. Promoting auto-pay methods could improve customer retention.

Monthly Charges Analysis

The relationship between monthly charges and churn was explored.

Findings:

- Customers paying higher monthly charges were more likely to churn.
- Lower monthly charge groups generally demonstrated better retention.

Business Insight:

Price sensitivity plays an important role in customer decisions. Customers with higher bills may require additional value-added services or discounts to justify the cost.

Internet Service Analysis

Internet service categories included:

- DSL
- Fiber Optic
- No Internet Service

Findings:

- Fiber Optic customers showed the highest churn levels.
- DSL customers had comparatively lower churn.
- Customers without internet service showed the lowest churn.

Business Insight:

The unexpectedly high churn among Fiber Optic customers may indicate pricing concerns, service quality issues, or unmet customer expectations. Further investigation is recommended.

Service-Based Churn Analysis

Several value-added services were analyzed:

- Online Security
- Online Backup
- Device Protection
- Tech Support
- Streaming TV

- Streaming Movies

Key Findings:

Online Security

- Customers without Online Security had significantly higher churn.
- Customers using Online Security were more likely to remain.

Tech Support

- Similar to Online Security, customers without Tech Support churned at much higher rates.

Online Backup & Device Protection

- Customers subscribing to these services exhibited somewhat lower churn.

Streaming Services

- Streaming TV and Streaming Movies showed moderate impact on retention.
- Churn differences were smaller compared to Security and Tech Support services.

Business Insight:

Security and support services appear to create the strongest customer stickiness. Customers using these services become more integrated into the ecosystem and are less likely to leave.

4. Major Drivers of Churn

Based on the analysis, the strongest indicators of churn are:

1. Month-to-Month Contract
2. Low Customer Tenure
3. Fiber Optic Internet Service
4. Lack of Online Security
5. Lack of Tech Support
6. Higher Monthly Charges
7. Electronic Check Payment Method
8. Senior Citizen Status

These variables should be prioritized when building churn prediction models.

5. Business Recommendations

Recommendation 1: Focus on New Customers

Implement onboarding programs, welcome offers, and early engagement campaigns during the first six months.

Recommendation 2: Promote Long-Term Contracts

Offer discounts and incentives for customers switching to annual or multi-year contracts.

Recommendation 3: Bundle Security Services

Provide bundled packages that include Online Security, Backup, and Tech Support services.

Recommendation 4: Investigate Fiber Optic Churn

Conduct customer surveys and service quality assessments to understand why Fiber Optic users are leaving.

Recommendation 5: Optimize Pricing Strategy

Review pricing for customers with high monthly charges and consider loyalty discounts.

Recommendation 6: Improve Payment Experience

Encourage migration from Electronic Check to automated payment methods.

Recommendation 7: Create Senior Citizen Retention Programs

Offer personalized support and simplified service packages for senior customers.

6. Conclusion

This exploratory analysis successfully identified several factors strongly associated with customer churn. The results suggest that churn is primarily driven by customer tenure, contract type, internet service choice, support-related services, pricing, and payment behavior. Among all variables analyzed, **month-to-month contracts, lack of Online Security and Tech Support, Fiber Optic internet service, and low tenure emerged as the most significant churn indicators.**